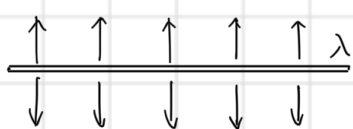
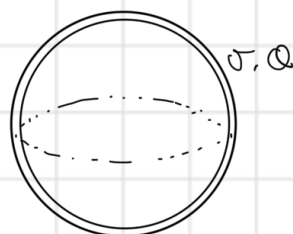


Recall.

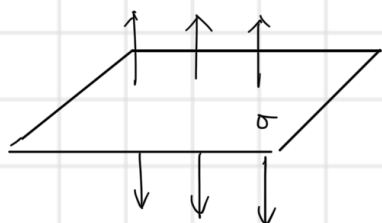
Electric field $\mathbf{E} \Rightarrow \mathbf{F} = q \cdot \mathbf{E}$



$$E = \frac{\lambda}{2\pi\epsilon_0 r}, \quad k = \frac{1}{4\pi\epsilon_0}$$



$$E = \begin{cases} 0 & \text{inside} \\ \frac{Q}{4\pi\epsilon_0 r^2} & \text{outside} \end{cases}$$



$$E = \frac{\sigma}{2\epsilon_0}$$

"Symmetry".

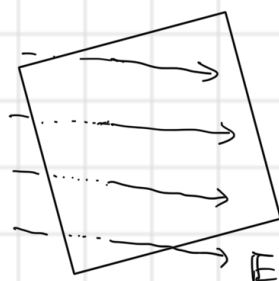
Electric flux

Φ_E measures how much electric field flows through a given surface. How to define it?

- if $\mathbf{E} = \text{const}$, and perpendicular to the flat surface:

$$\Phi_E = \mathbf{E} \cdot \mathbf{A}$$

area of the surface.



- if $\mathbf{E} = \text{const}$, surface is flat but $\mathbf{E}$ is not perpendicular to the surface:



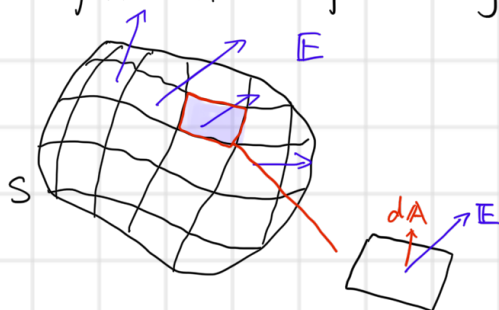
$$\Phi_E = EA \cos \theta$$

unit normal vector to the surface.

Introduce $\mathbf{A} := A \hat{n}$

$$\Rightarrow \Phi_E = \mathbf{E} \cdot \mathbf{A}$$

- In general, surface may not be flat, $\mathbf{E}$ may not be constant.



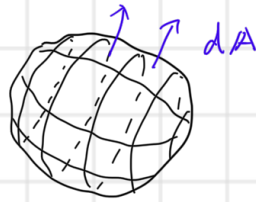
$$\Phi_E = \int_S d\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{A}$$

Remark.

the direction of $d\mathbf{A}$ needs to be decided.

$$d\Phi_E = \mathbf{E} \cdot d\mathbf{A}$$

Standard choice for closed surfaces:

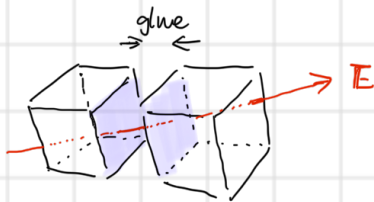


point outward

Gauss's Law

Let S be a closed surface, $\partial S = \emptyset$.

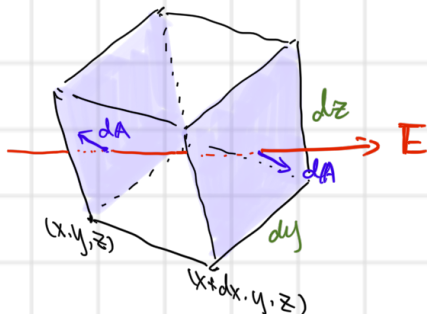
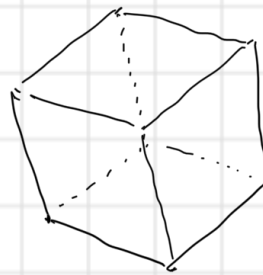
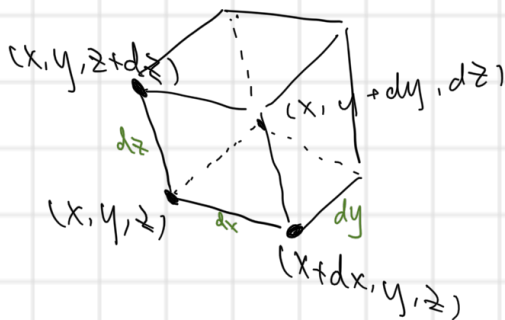
To calculate Φ_E , we approximate the region bounded by S by a collection of cubes.



Fluxes through the colored faces cancel each other.

$\Rightarrow$ Glued faces don't contribute to the total flux, only external ones do.

$\Rightarrow \Phi_E$ through $S = \sum_{\text{all cubes}} \Phi_E$ through cubes' surfaces.



$$-E_x(x, y, z) dy dz$$

$$+E_x(x+dx, y, z) dy dz$$

Φ_E through this pair

$$= [E_x(x+dx, y, z) - E_x(x, y, z)] dy dz$$

$$= \frac{\partial E_x}{\partial x}(x, y, z) dx dy dz$$

$$= \frac{\partial E_x}{\partial x}(x, y, z) dV.$$

$\Rightarrow \Phi_E$ through the cube

$$= \left[\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \right] (x, y, z) \cdot dV.$$

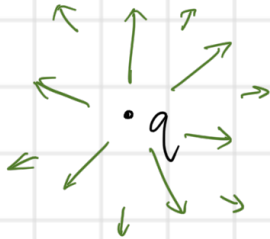
divergence of $\mathbf{E} =: \nabla \cdot \mathbf{E}$, $\text{div } \mathbf{E}$

$$\nabla := i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z}$$

$$\Rightarrow \Phi_E \text{ through } S = \int_V \nabla \cdot \mathbf{E} \, dV.$$

For any vector field $\mathbf{V}(x, y, z)$, $\int_S \mathbf{V} \cdot d\mathbf{A} = \int_V (\nabla \cdot \mathbf{V}) \, dV$
 "Gauss' theorem"

Let's consider a point charge q , to apply this result.



$$\mathbf{E}(r) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \hat{r}, \quad r \neq 0.$$

$$\begin{aligned} \nabla \cdot \left(\frac{\mathbf{r}}{r^3} \right) &= \frac{\partial}{\partial x} \left(\frac{x}{[x^2+y^2+z^2]^{3/2}} \right) + \dots + \dots \\ &= \sum_{\text{cyc}} \frac{1}{[x^2+y^2+z^2]^{3/2}} - \frac{3}{2} \frac{2x^2}{[x^2+y^2+z^2]^{5/2}} \\ &= \frac{3}{[x^2+y^2+z^2]^{3/2}} - \frac{3}{[x^2+y^2+z^2]^{3/2}} = 0 \end{aligned}$$

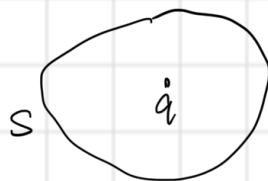
$\Rightarrow$ The divergence $\nabla \cdot \mathbf{E} = 0$ everywhere except $r=0$.

If S contains q ,

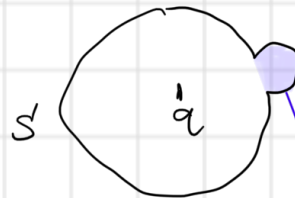
$$\Phi_E \text{ through } S = \int_V \nabla \cdot \mathbf{E} \, dV$$

In this case $\Phi_E \neq 0$.

To calculate Φ_E , we use the fact Φ_E doesn't depend on the shape of S' . Because if we change the shape of S ,



$$\Phi_E(S)$$

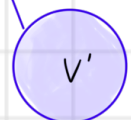


$$\Phi_E(S') = \Phi_E(S) + \Phi_E(S' - S).$$

S doesn't contain q

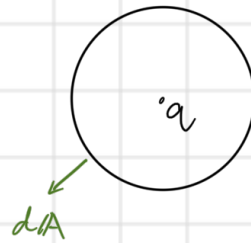
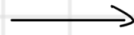
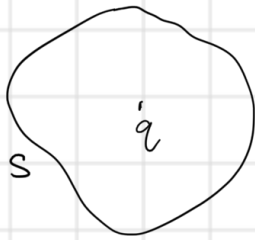
$\Rightarrow \Phi_E \text{ through } S = 0.$

$$\begin{aligned} \leftarrow \Phi_E(S' - S) &= \int_{V'} \nabla \cdot \mathbf{E} \, dV \\ &= 0 \end{aligned}$$



$\Rightarrow$ we can change S and Φ_E is still the same as long as the surface contains q .

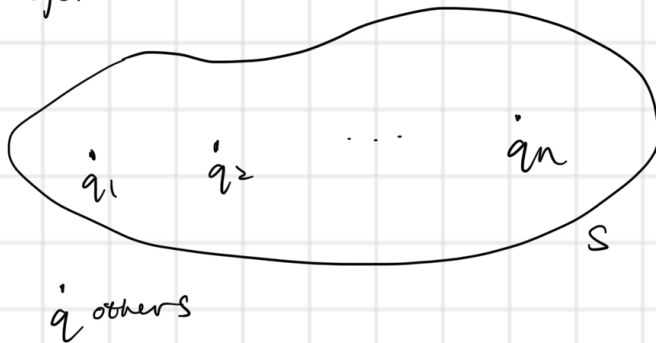
$\Rightarrow$



$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

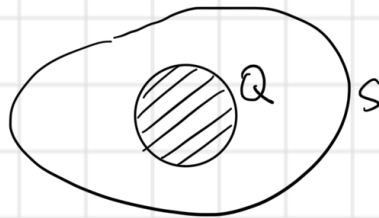
$$\Phi_E = \int_S E \cdot dA = 4\pi r^2 \cdot \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} = \boxed{\frac{q}{\epsilon_0}}$$

In general:



$$\Phi_E = \frac{1}{\epsilon_0} \sum_{i=1}^n q_i$$

Continuously:

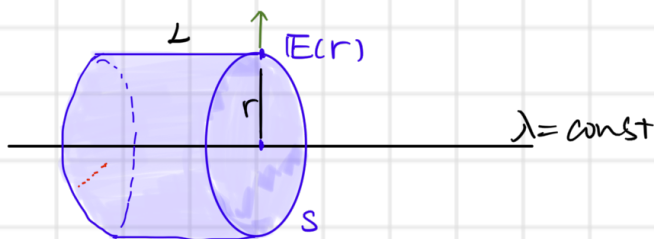


$$\Phi_E = \frac{Q}{\epsilon_0}$$

Gauss Law:

$$\Phi_E = \frac{\text{total charge contained in } S}{\epsilon_0}$$

Ex. infinite line of charge.



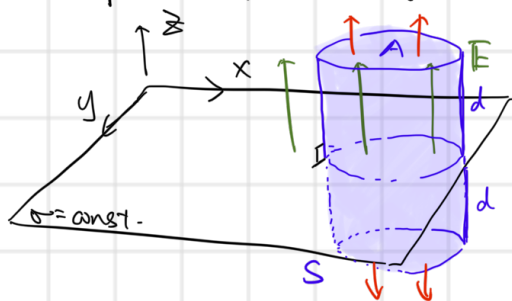
By translation & rotation symmetry & $z \mapsto -z$ symmetry, we deduce $\vec{E}(r) = E(r) \cdot \hat{r}$

$$\begin{aligned} \Phi_E(S) &= E(r) \cdot A_{\text{cylinder side surface}} \\ &= E(r) 2\pi r L \end{aligned}$$

Gauss' Law:

$$\Phi_E(S) = \frac{Q}{\epsilon_0} = \frac{\lambda \cdot L}{\epsilon_0} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{aligned} E(r) &= \frac{\lambda}{2\pi\epsilon_0 r} \\ E(r) &= \frac{\lambda \hat{r}}{2\pi\epsilon_0 r} \end{aligned}$$

Ex. Infinite plane of charges.



By x - y translation symmetry

$$\Rightarrow E = E(z).$$

$z \mapsto -z$ symmetry $\Rightarrow E = E(|z|).$

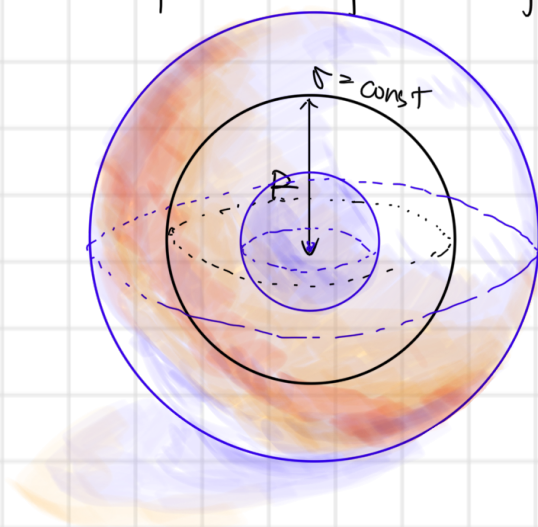
x - y Rotational symmetry $\Rightarrow$ directed along z -axis.

$$E(z) = \begin{cases} E(|z|) \hat{k}, & z > 0 \\ -E(|z|) \hat{k}, & z < 0. \end{cases}$$

$$\begin{aligned} \Phi_E &= 2 \cdot A \cdot E(|z|) \\ &= \frac{Q}{\epsilon_0} = \frac{\sigma \cdot A}{\epsilon_0} \end{aligned}$$

$$\Rightarrow E(|z|) = \frac{\sigma}{2\epsilon_0}$$

Ex. Sphere of charge



Spherical symmetry $\Rightarrow E = E(r) \hat{r}$

$$\Phi_E = E(r) \cdot 4\pi L^2$$

$$= \frac{1}{\epsilon_0} \begin{cases} 4\pi R^2 \cdot \sigma & (L > R) \\ 0 & (L < R) \end{cases}$$

$$\Rightarrow E(r=L) = \begin{cases} \frac{R^2 \sigma}{\epsilon_0 L^2} = \frac{Q}{4\pi \epsilon_0 L^2}, & L > R, \\ 0, & L < R. \end{cases}$$

$$\boxed{\nabla \cdot E = \frac{\rho}{\epsilon_0}}$$

maxwell 1/4